1 Vector partitions and numerical functions

Let $\mathbb{N}$ denote the nonnegative integers and $\mathbb{P}$ the positive integers.

Let $\mathbf{u} \in \mathbb{N}^r$. Write $\mathbf{u} = (u_1, \ldots, u_r)$ and $\mathrm{wt}(\mathbf{u}) = \sum u_i$. A **vector partition** of $\mathbf{u}$ is a multiset of nonzero vectors $\lambda = \lambda^{(1)}, \ldots, \lambda^{(k)} \in \mathbb{N}^r$ that sum to $\mathbf{u}$.

- The **length** is $\ell(\lambda) = k$.
- The **weight** $wt(\lambda)$ is the sum of all entries in its vectors (so $\mathrm{wt}(\lambda) = \mathrm{wt}(\mathbf{u})$).
- The **sign** is $\mathrm{sign}(\lambda) = (-1)^{\text{number of vectors with even sum}}$
- For a vector partition λ in which each vector v_i occurs with multiplicity m_i, define

$$|\lambda| = \prod_i m_i! \qquad \text{and} \qquad \lambda! = \prod_i \prod_{x \in v_i} (x!)^{m_i}.$$

For example, the vector partitions of $\mathbf{u} = (2, 1) = 21$ are as follows.

λ	$\mathrm{sign}(\lambda)$	$\lvert\lambda\rvert$	$\lambda!$
21	$+1$	1	2
$20, 01$	-1	1	2
$10, 11$	-1	1	1
$10, 10, 01$	$+1$	2	1

A vector partition is **unitary** if it is a partition of the all-ones vector $\mathbf{1} \in \mathbb{N}^k$. There is an obvious bijection between unitary vector partitions and set partitions of $[k]$.

The **type** of a set partition $\pi = B_1 | \cdots | B_\ell$, with respect to a vector $\mathbf{u} = (u_1, \ldots, u_r)$ of weight n, is the vector partition $\mathrm{type}_{\mathbf{u}}(\pi) = \lambda = \lambda_1 \cdots \lambda_\ell$ where λ_k is the vector in $\mathbb{N}^r$ whose ith coordinate is

$$\#\{j \in B_k \mid u_1 + \cdots + u_{i-1} < j \leq i_1 + \cdots + u_i\}.$$

the number of elements of B_k such that in the ith equivalence class.

For a vector $\mathbf{u}$ of weight n and a vector partition λ of weight n, the number of set partitions π of type $\mathrm{type}_{\mathbf{u}}(\pi)$ is

$$\binom{\mathbf{u}}{\lambda} := \frac{\mathbf{u}!}{\lambda!\,|\lambda|}.$$

2 MacMahon symmetric functions

Fix $r \in \mathbb{P}$ and let $\mathcal{A} = \{x_{i,j} \mid i \in [r],\ j \in \mathbb{P}\}$. Everything that follows takes place in the ring of formal power series $R = \Bbbk[[\mathcal{A}]]$, where $\Bbbk$ is a field (assumed to be of characteristic 0 if necessary). There is an $\mathbb{N}^r$-grading on R; the homogeneous component of degree $\mathbf{u} = (u_1, \ldots, u_r) \in \mathbb{N}^r$ is the vector space span of monomials having total degree u_i in the variables $\{x_{i,j} \mid j \in \mathbb{P}\}$.

The group $\mathfrak{S}$ of permutations of $\mathbb{P}$ acts diagonally on $\mathcal{A}$ by $\sigma(x_{i,j}) = x_{i,\sigma(j)}$; this action extends to R. The invariants of the action are the **MacMahon (symmetric) functions**. They form a graded subring $\mathfrak{M} \subseteq R$, with $\mathfrak{M}_{\mathbf{u}} = \mathfrak{M} \cap R_{\mathbf{u}}$.

Following Rosas [Ros01], the standard combinatorial bases for $\mathfrak{M}$ are as follows. For $\mathbf{u} \in \mathbb{N}^r$, each basis for $\mathfrak{M}_{\mathbf{u}}$ is indexed by the vector partitions of $\mathbf{u}$.

Let $\lambda = \{\lambda^{(1)}, \ldots, \lambda^{(k)}\}$ be a vector partition, where $\lambda^{(j)} = (\ell_{1,j}, \ldots, \ell_{r,j})$. The **monomial MacMahon function** m_λ is the sum of all monomials in the $\mathfrak{S}$-orbit of the monomial

$$\prod_{j=1}^{k} \prod_{i=1}^{r} x_{i,j}^{\ell_{i,j}}.$$

For $\mathbf{a} = (a_1, \ldots, a_r) \in \mathbb{N}^r$, the **power-sum MacMahon function** is

$$p_{\mathbf{a}} = \sum_{i=1}^{\infty} x_{1,i}^{a_1} x_{2,i}^{a_2} \cdots x_{r,i}^{a_r};$$

the **elementary MacMahon function** is

$$e_{\mathbf{a}} = [t_1^{a_1} \cdots t_r^{a_r}] \prod_{i=1}^{\infty} 1 + t_1 x_{1,i} + t_2 x_{2,i} + \cdots + t_r x_{r,i}$$

and the **complete homogeneous MacMahon function** is

$$h_{\mathbf{a}} = [t_1^{a_1} \cdots t_r^{a_r}] \prod_{i=1}^{\infty} \frac{1}{1 - t_1 x_{1,i} - t_2 x_{2,i} - \cdots - t_r x_{r,i}}.$$

These are all multiplicative bases: for a vector partition $\lambda = \{\lambda^{(1)}, \ldots, \lambda^{(k)}\}$ we set

$$p_\lambda = \prod_{j=1}^{k} p_{\lambda^{(j)}}, \qquad e_\lambda = \prod_{j=1}^{k} e_{\lambda^{(j)}}, \qquad h_\lambda = \prod_{j=1}^{k} h_{\lambda^{(j)}}.$$

Finally, there is an involutive automorphism $\omega \colon \mathfrak{M} \to \mathfrak{M}$ defined by $\omega(e_\lambda) = h_\lambda$. The **forgotten MacMahon functions** are defined by $f_\lambda = \operatorname{sign}(\lambda) m_\lambda$.

The unitary MacMahon symmetric functions are the graded piece $\mathfrak{M}_{(1)^r}$. These are the focus of [Dou72].

3 Transitions between bases

First, a standard linear algebra fact. Let V be a vector space of finite dimension n over ground field $\Bbbk$

Proposition 1. *Suppose that $B = \{v_1, \ldots, v_n\}$ and $B^* = \{v_1^*, \ldots, v_n^*\}$ are bases for V that are orthogonal, i.e.,*

$$\langle v_i, v_j^* \rangle = \delta_{ij} k_i$$

for some nonzero constants $k_1, \ldots, k_n$. Then the unique expansion of u with respect to B is

$$u = \sum_{i=1}^{n} \frac{\langle v_i^*, u \rangle}{k_i} v_i. \tag{1}$$

Proof. For every $j \in [n]$ we have

$$\left\langle v_j^*, \sum_{i=1}^{n} \frac{\langle v_i^*, u \rangle}{k_i} v_i \right\rangle = \sum_{i=1}^{n} \frac{\langle v_i^*, u \rangle}{k_i} \langle v_j^*, v_i \rangle = \sum_{i=1}^{n} \langle v_i^*, u \rangle \delta_{ij} = \langle v_j^*, u \rangle.$$

That is, every element of B^* has the same inner product with the left and right-hand sides of (1). This is necessary and sufficient for equality. $\square$

This proposition reduces the problem of expanding a vector in a particular basis B to computing inner products with an orthogonal basis B^*. (The best possible case is that $k_i = 1$ for all i, i.e., B^* is the dual basis to B. However, we're not always that lucky.)

Define a scalar product on $\mathfrak{M}$:

$$\boxed{\langle h_\lambda,\, m_\mu \rangle = \delta_{\lambda\mu}.} \tag{2}$$

The idea is to use this for basis transitions, along the lines discussed earlier. Specifically, formula (1) says that for any $F \in \mathfrak{M}_{\mathbf{u}}$, we have

$$F = \sum_{\lambda \vdash u} \langle h_\lambda,\, F \rangle m_\lambda \qquad \text{and} \qquad F = \sum_{\lambda \vdash u} \langle m_\lambda,\, F \rangle h_\lambda. \tag{3}$$

Thus we have to know how to compute inner products.

There is a "lifting map" $\hat{\rho}$ [Ros01, Defn. 6] sending MacMahon functions to unitary (Doubilet) symmetric functions. Specifically:

$$\binom{\mathbf{u}}{\lambda} |\lambda|\, m_\lambda \xmapsto{\hat{\rho}} \sum_{\pi\,:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda} m_\pi \tag{4a}$$

$$\binom{\mathbf{u}}{\lambda} \lambda!\, h_\lambda \xmapsto{\hat{\rho}} \sum_{\pi\,:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda} h_\pi \tag{4b}$$

$$\binom{\mathbf{u}}{\lambda} \lambda!\, e_\lambda \xmapsto{\hat{\rho}} \sum_{\pi\,:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda} e_\pi \tag{4c}$$

$$\binom{\mathbf{u}}{\lambda} p_\lambda \xmapsto{\hat{\rho}} \sum_{\pi\,:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda} p_\pi \tag{4d}$$

$$\binom{\mathbf{u}}{\lambda} |\lambda|\, f_\lambda \xmapsto{\hat{\rho}} \sum_{\pi\,:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda} f_\pi \tag{4e}$$

Moreover, for all $f, g \in \mathfrak{M}_{\mathbf{u}}$ we have [Ros01, Prop. 7]

$$\langle f,\, g \rangle = \mathbf{u}!\, \langle \hat{\rho}(f),\, \hat{\rho}(g) \rangle \tag{5}$$

Prop. 1 together with the scalar product (2) says in particular that for $F \in \mathfrak{M}_{\mathbf{u}}$ we have

$$F = \sum_{\lambda \vdash \mathbf{u}} \langle m_\lambda,\, F \rangle\, h_\lambda, \qquad\qquad F = \sum_{\lambda \vdash \mathbf{u}} \langle h_\lambda,\, F \rangle\, m_\lambda. \tag{6}$$

Doubilet [Dou72, Appendix 2] calculated the scalar products for all five families of unitary symmetric functions (which require things like lattice operations and Möbius and zeta functions in the set partition lattice Π_n). Therefore, we should be able to compute scalar products, and thus basis expansions, for all MacMahon functions. Warning: Doubilet's notation for the standard bases is not the usual modern notation (he used k, a, h, s, f for what we call m, e, h, p, f — perhaps this was not standardized in 1970?), and each of his formulas has an extra $n!$ that needs to be deleted (he was working in an analogous ring where the $n!$ makes sense). Rosas [Ros01] reproduced Doubilet's calculations (in modern notation and without the $n!$'s that are extraneous in our context).

4 Expansions in the m-basis

$$
\begin{aligned}
p_\mu &= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \langle \hat{\rho}(h_\lambda),\, \hat{\rho}(p_\mu) \rangle\, m_\lambda && \text{(by (3) and (5))} \\
&= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \left\langle \frac{1}{\binom{\mathbf{u}}{\lambda} \lambda!} \sum_{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda} h_\pi,\ \frac{1}{\binom{\mathbf{u}}{\mu}} \sum_{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu} p_\sigma \right\rangle m_\lambda && \text{(by (4b) and (4d))} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu}} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda} \lambda!} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu}} \langle h_\pi,\, p_\sigma \rangle\, m_\lambda && \text{(by bilinearity)} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu}} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda} \lambda!} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu}} \zeta(\sigma, \pi) m_\lambda && \text{(by Doubilet formula \#3)} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu}} \sum_{\lambda \vdash \mathbf{u}} \frac{\#\{(\pi, \sigma) \mid \mathrm{type}_{\mathbf{u}}(\pi) = \lambda,\ \mathrm{type}_{\mathbf{u}}(\sigma) = \mu,\ \sigma \leq \pi\}}{\binom{\mathbf{u}}{\lambda} \lambda!} m_\lambda
\end{aligned}
$$

$$
\begin{aligned}
h_\mu &= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \langle \hat{\rho}(h_\lambda),\, \hat{\rho}(h_\mu) \rangle\, m_\lambda && \text{(by (3) and (5))} \\
&= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \left\langle \frac{1}{\binom{\mathbf{u}}{\lambda} \lambda!} \sum_{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda} h_\pi,\ \frac{1}{\binom{\mathbf{u}}{\mu} \mu!} \sum_{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu} h_\sigma \right\rangle m_\lambda && \text{(by (4b) twice)} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu} \mu!} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda} \lambda!} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu}} \langle h_\pi,\, h_\sigma \rangle\, m_\lambda && \text{(by bilinearity)} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu} \mu!} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda} \lambda!} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu}} \lambda(\sigma \wedge \pi)!\, m_\lambda && \text{(by Doubilet formula \#9)}
\end{aligned}
$$

$$
\begin{aligned}
e_\mu &= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \langle \hat{\rho}(h_\lambda),\, \hat{\rho}(e_\mu) \rangle\, m_\lambda && \text{(by (3) and (5))} \\
&= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \left\langle \frac{1}{\binom{\mathbf{u}}{\lambda} \lambda!} \sum_{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda} h_\pi,\ \frac{1}{\binom{\mathbf{u}}{\mu} \mu!} \sum_{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu} e_\sigma \right\rangle m_\lambda && \text{(by (4b) and (4c))} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu} \mu!} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda} \lambda!} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu}} \langle h_\pi,\, e_\sigma \rangle\, m_\lambda && \text{(by bilinearity)} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu} \mu!} \sum_{\lambda \vdash \mathbf{u}} \frac{\#\{(\pi, \sigma) \mid \mathrm{type}_{\mathbf{u}}(\pi) = \lambda,\ \mathrm{type}_{\mathbf{u}}(\sigma) = \mu,\ \pi \wedge \sigma = \hat{\mathbf{0}}\}}{\binom{\mathbf{u}}{\lambda} \lambda!} m_\lambda && \text{(by Doubilet formula \#6)}
\end{aligned}
$$

$$
\begin{aligned}
f_\mu &= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \langle \hat{\rho}(h_\lambda),\, \hat{\rho}(f_\mu) \rangle\, m_\lambda && \text{(by (3) and (5))} \\
&= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \left\langle \frac{1}{\binom{\mathbf{u}}{\lambda}\lambda!} \sum_{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda} h_\pi,\ \frac{1}{\binom{\mathbf{u}}{\mu}|\mu|} \sum_{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu} f_\sigma \right\rangle m_\lambda && \text{(by (4b) and (4d))} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu}|\mu|} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda}\lambda!} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu}} \langle h_\pi,\, f_\sigma \rangle\, m_\lambda && \text{(by bilinearity)} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu}|\mu|} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda}\lambda!} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu \\ \sigma \leq \pi}} \lambda(\sigma,\pi)! m_\lambda && \text{(by Doubilet formula \#13)}
\end{aligned}
$$

5 Expansions in the h-basis

$$
\begin{aligned}
m_\mu &= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \langle \hat{\rho}(m_\lambda),\, \hat{\rho}(m_\mu) \rangle\, h_\lambda && \text{(by (3) and (5))} \\
&= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \left\langle \frac{1}{\binom{\mathbf{u}}{\lambda}|\lambda|} \sum_{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda} m_\pi,\ \frac{1}{\binom{\mathbf{u}}{\mu}|\mu|} \sum_{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu} m_\sigma \right\rangle h_\lambda && \text{(by (4a) twice)} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu}|\mu|} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda}|\lambda|} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu}} \langle m_\pi,\, m_\sigma \rangle\, h_\lambda && \text{(by bilinearity)} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu}|\mu|} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda}|\lambda|} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu}} \sum_{\tau \geq \pi \vee \sigma} \frac{\mu(\pi,\tau)\mu(\sigma,\tau)}{\mu(\hat{\mathbf{0}},\tau)}\, h_\lambda && \text{(by Doubilet formula \#7)}
\end{aligned}
$$

$$
\begin{aligned}
p_\mu &= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \langle \hat{\rho}(m_\lambda),\, \hat{\rho}(p_\mu) \rangle\, h_\lambda && \text{(by (3) and (5))} \\
&= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \left\langle \frac{1}{\binom{\mathbf{u}}{\lambda}|\lambda|} \sum_{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda} m_\pi,\ \frac{1}{\binom{\mathbf{u}}{\mu}} \sum_{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu} p_\sigma \right\rangle h_\lambda && \text{(by (4a) and (4d))} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu}} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda}|\lambda|} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu}} \langle m_\pi,\, p_\sigma \rangle\, h_\lambda && \text{(by bilinearity)} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu}} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda}|\lambda|} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu \\ \pi \leq \sigma}} \frac{\mu(\pi,\sigma)}{|\mu(\hat{\mathbf{0}},\sigma)|}\, h_\lambda && \text{(by Doubilet formula \#10)}
\end{aligned}
$$

$$
\begin{aligned}
e_\mu &= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \langle \hat{\rho}(m_\lambda),\, \hat{\rho}(e_\mu) \rangle\, h_\lambda && \text{(by (3) and (5))} \\
&= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \left\langle \frac{1}{\binom{\mathbf{u}}{\lambda}|\lambda|} \sum_{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda} m_\pi,\ \frac{1}{\binom{\mathbf{u}}{\mu}\mu!} \sum_{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu} e_\sigma \right\rangle h_\lambda && \text{(by (4a) and (4c))} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu}\mu!} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda}|\lambda|} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu}} \langle m_\pi,\ e_\sigma \rangle\ h_\lambda && \text{(by bilinearity)} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu}\mu!} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda}|\lambda|} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu \\ \pi \le \sigma}} \mathrm{sign}(\pi)\lambda(\pi,\sigma)!\, h_\lambda && \text{(by Doubilet formula \#8)}
\end{aligned}
$$

$$
\begin{aligned}
f_\mu &= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \langle \hat{\rho}(m_\lambda),\, \hat{\rho}(f_\mu) \rangle\, h_\lambda && \text{(by (3) and (5))} \\
&= \mathbf{u}! \sum_{\lambda \vdash \mathbf{u}} \left\langle \frac{1}{\binom{\mathbf{u}}{\lambda}|\lambda|} \sum_{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda} m_\pi,\ \frac{1}{\binom{\mathbf{u}}{\mu}|\mu|} \sum_{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu} f_\sigma \right\rangle h_\lambda && \text{(by (4a) and (4e))} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu}|\mu|} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda}|\lambda|} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu}} \langle m_\pi,\ f_\sigma \rangle\ h_\lambda && \text{(by bilinearity)} \\
&= \frac{\mathbf{u}!}{\binom{\mathbf{u}}{\mu}|\mu|} \sum_{\lambda \vdash \mathbf{u}} \frac{1}{\binom{\mathbf{u}}{\lambda}|\lambda|} \sum_{\substack{\pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\lambda \\ \sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\mu}} \sum_{\tau \ge \sigma \vee \pi} \frac{|\mu(\sigma,\tau)| \cdot \mu(\pi,\tau)}{\mu(\hat{\mathbf{0}},\tau)}\, h_\lambda && \text{(by Doubilet formula \#11)}
\end{aligned}
$$

6 Expansions in the p-basis

Let $\mathbf{u} \in \mathbb{N}^r$ and $n- = \mathrm{wt}(\mathbf{u})$. Rosas [Ros01, p.333] states that for $\lambda \vdash \mathbf{u}$,

$$
\langle p_\lambda,\, p_\mu \rangle = \frac{\delta_{\lambda,\mu} |\lambda|\, \lambda!}{|\mu(\hat{\mathbf{0}}, \lambda)|} \tag{7}
$$

where $\mu(\hat{\mathbf{0}}, \lambda)$ means $\mu(\hat{\mathbf{0}}, \pi)$ for any set partition π such that $\mathrm{type}_u(\pi) = \lambda$. This can be verified using the lifting maps and Doubilet's formula. For the purpose of computation, a set partition π can be constructed as follows:

- Partition $[n]$ into sets $A_1, \ldots, A_r$, with $|A_i| = u_i$.
- For each vector $\lambda^{(i)} = (\lambda_1^{(i)}, \ldots, \lambda_r^{(i)}) \in \lambda$, make a new block B_i consisting of the $\lambda_j^{(i)}$ smallest elements of A_j, for each $j \in [r]$. Remove those elements from A_j when adding them to B_i.
- At this point the sets A_j should all be empty, and the desired set partition is $\pi = \{B_1, \ldots, B_{\ell(\lambda)}\}$.

Let $A_1 = [1, u_1]$, $A_2 = [u_1 + 1, u_1 + u_2], \ldots A_k = [u_1 + \cdots + u_{k-1} + 1, u_1 + u_2], \ldots A_2 = [u_1 + 1, u_1 + u_2], \ldots$

Therefore, Proposition 1 says that for every $F \in \mathfrak{M}$,

$$F = \sum_{\lambda} \frac{\langle p_\lambda,\, F\rangle |\mu(\hat{\mathbf{0}}, \lambda)|}{|\lambda|\, \lambda!} p_\lambda = \sum_{\lambda} \frac{\langle p_\lambda,\, F\rangle |\mu(\hat{\mathbf{0}}, \lambda)| \binom{\mathbf{u}}{\lambda}}{\mathbf{u}!} p_\lambda \tag{8}$$

So, as before:

$$\begin{aligned}
m_\mu &= \sum_{\lambda} \left\langle \binom{\mathbf{u}}{\lambda} \hat{\rho}(p_\lambda),\, \hat{\rho}(m_\mu) \right\rangle |\mu(\hat{\mathbf{0}}, \lambda)| p_\lambda && \text{(by (8) and (5))} \\
&= \frac{1}{\binom{\mathbf{u}}{\mu} |\mu|} \sum_{\lambda} |\mu(\hat{\mathbf{0}}, \lambda)| \sum_{\substack{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\lambda \\ \pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\mu}} \langle p_\sigma,\, m_\pi \rangle\, p_\lambda && \text{(by (4d) and (4a) and bilinearity)} \\
&= \frac{1}{\binom{\mathbf{u}}{\mu} |\mu|} \sum_{\lambda} \left[|\mu(\hat{\mathbf{0}}, \lambda)| \sum_{\substack{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\lambda \\ \pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\mu \\ \pi \leq \sigma}} \frac{\mu(\pi, \sigma)}{|\mu(\hat{\mathbf{0}}, \sigma)|} \right] p_\lambda && \text{(by Doubilet formula \#10)}
\end{aligned}$$

$$\begin{aligned}
h_\mu &= \sum_{\lambda} \langle \hat{\rho}(p_\lambda),\, \hat{\rho}(h_\mu) \rangle |\mu(\hat{\mathbf{0}}, \lambda)| \binom{\mathbf{u}}{\lambda} p_\lambda && \text{(by (8) and (5))} \\
&= \frac{1}{\binom{\mathbf{u}}{\mu} \mu!} \sum_{\lambda} |\mu(\hat{\mathbf{0}}, \lambda)| \sum_{\substack{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\lambda \\ \pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\mu}} \langle p_\sigma,\, h_\pi \rangle\, p_\lambda && \text{(by (4d) and (4b) and bilinearity)} \\
&= \frac{1}{\binom{\mathbf{u}}{\mu} \mu!} \sum_{\lambda} \left[|\mu(\hat{\mathbf{0}}, \lambda)| \sum_{\substack{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\lambda \\ \pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\mu \\ \sigma \leq \pi}} \right] p_\lambda && \text{(by Doubilet formula \#3)}
\end{aligned}$$

$$\begin{aligned}
e_\mu &= \sum_{\lambda} \langle \hat{\rho}(p_\lambda),\, \hat{\rho}(e_\mu) \rangle |\mu(\hat{\mathbf{0}}, \lambda)| \binom{\mathbf{u}}{\lambda} p_\lambda && \text{(by (8) and (5))} \\
&= \frac{1}{\binom{\mathbf{u}}{\mu} \mu!} \sum_{\lambda} |\mu(\hat{\mathbf{0}}, \lambda)| \sum_{\substack{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\lambda \\ \pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\mu}} \langle p_\sigma,\, e_\pi \rangle\, p_\lambda && \text{(by (4d) and (4c) and bilinearity)} \\
&= \frac{1}{\binom{\mathbf{u}}{\mu} \mu!} \sum_{\lambda} \left[|\mu(\hat{\mathbf{0}}, \lambda)| \sum_{\substack{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\lambda \\ \pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\mu \\ \sigma \leq \pi}} \mathrm{sign}(\sigma) \right] p_\lambda && \text{(by Doubilet formula \#4)}
\end{aligned}$$

$$
\begin{aligned}
f_\mu &= \sum_\lambda \langle \hat\rho(p_\lambda),\ \hat\rho(f_\mu)\rangle |\mu(\hat{\mathbf{0}},\lambda)| \binom{\mathbf{u}}{\lambda} p_\lambda && \text{(by (8) and (5))} \\
&= \frac{1}{\binom{\mathbf{u}}{\mu}|\mu|} \sum_\lambda |\mu(\hat{\mathbf{0}},\lambda)| \sum_{\substack{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\lambda \\ \pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\mu}} \langle p_\sigma,\ f_\pi\rangle\, p_\lambda && \text{(by (4d) and (4e) and bilinearity)} \\
&= \frac{1}{\binom{\mathbf{u}}{\mu}|\mu|} \sum_\lambda \left[|\mu(\hat{\mathbf{0}},\lambda)| \sum_{\substack{\sigma:\ \mathrm{type}_{\mathbf{u}}(\sigma)=\lambda \\ \pi:\ \mathrm{type}_{\mathbf{u}}(\pi)=\mu \\ \pi\le\sigma}} \mathrm{sign}(\pi)\,\mathrm{sign}(\sigma)\,\frac{\mu(\pi,\sigma)}{|\mu(\hat{\mathbf{0}},\sigma)|} \right] p_\lambda && \text{(by Doubilet formula \#14)}
\end{aligned}
$$

7 The omega involution, and expansions in the e- and f-bases

There is an involutive automorphism $\omega\colon \mathfrak{M} \to \mathfrak{M}$ (called θ in [Dou72]) defined by $\omega(e_\lambda) = h_\lambda$. Rosas defines it at [Ros01, p.328] and then uses it to define the forgotten basis as $f_\lambda = \mathrm{sign}(\lambda)\omega(m_\lambda)$. Rosas also states as a fact (in the middle of a proof on p.334) that $\omega(p_\lambda) = \mathrm{sign}(\lambda)p_\lambda$, although she does not explain where that formula comes from. It may be in Doublet somewhere. It certainly feels true (it's true for symmetric functions). In summary, ω acts on basis elements by

$$
\begin{aligned}
\omega(e_\lambda) &= h_\lambda, \\
\omega(h_\lambda) &= e_\lambda, \\
\omega(m_\lambda) &= \mathrm{sign}(\lambda) f_\lambda, \\
\omega(f_\lambda) &= \mathrm{sign}(\lambda) m_\lambda, \\
\omega(p_\lambda) &= \mathrm{sign}(\lambda) p_\lambda.
\end{aligned}
$$

References

[Dou72] Peter Doubilet, *On the foundations of combinatorial theory. VII. Symmetric functions through the theory of distribution and occupancy*, Studies in Appl. Math. **51** (1972), 377–396. MR 429577 2, 3, 8

[Ros01] Mercedes H. Rosas, *MacMahon symmetric functions, the partition lattice, and Young subgroups*, J. Combin. Theory Ser. A **96** (2001), no. 2, 326–340. MR 1864127 1, 3, 6, 8